(a)Estimate the probability that a student who studies for 40 h and has an undergrad GPA of 3.5 gets an A in the class.

Linear predictor: $\eta = -6 + 0.05(40) + 1(3.5) = -6 + 2 + 3.5 = -0.5$

Convert to probability using logistic function $p = e^{\eta} / (1 + e^{\eta})$:

$p = e^{(-0.5)} / (1 + e^{(-0.5)})$
$\approx 1 / (1 + e^{(0.5)}) \approx 1 / 2.6487 \approx$ **0.3775**

(b) How many hours would the student in part (a) need to study to have a 50 % chance of getting an A in the class?

For $p = 0.5 \rightarrow \text{logit}(0.5) = 0$.
Solve for X1:

$0 = -6 + 0.05X1 + 1(3.5)$
$\rightarrow 0.05X1 = 2.5 \rightarrow X1 = 2.5 / 0.05 =$ **50 hours**